

# 1. Team Details

Team name: bunker\_backer\_tmp

Members (name -- role):

S Sahith Somasundar -- modelling

Gara Shanmukh Sai -- data & augmentation

Navadeep Saran Y -- inference optimization

Vanisha Nadimpalli -- evaluation & docs

College: VNR Vignana Jyothi Institute of Engineering and Technology

Contact: navadeep.saran@gmail.com

## 2. Problem Statement Addressed

AI-Based Restoration of Degraded Images for Semiconductor Inspection (KLA PS01). A single lost pixel or noisy region in an inspection image can hide a real defect and cost a die -- restoration quality is a correctness problem, not a cosmetic one.

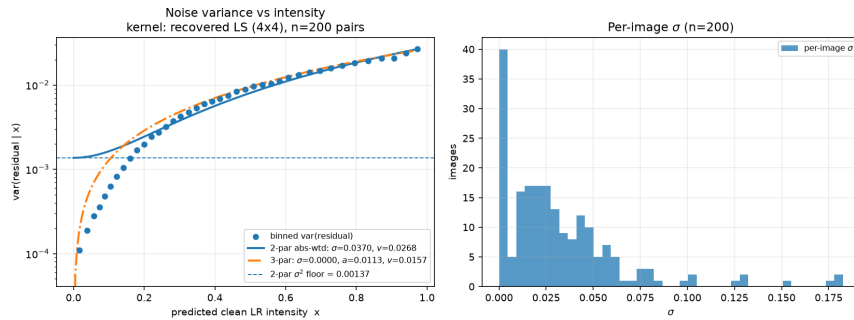
Three degradation mechanisms, applied in combination:

1. Speckle noise (signal-dependent, multiplicative-like variance term)
2. Additive/shot noise (signal-dependent linear term)
3. Downsampling (x2 decimation, GT/LR pair)

The pipeline must invert all three jointly, in whatever order they were applied, and generalise to content the training set never showed it.

### 3. Idea Description

The released dataset is 3200 training pairs and 400 test inputs of grayscale natural photographs, not semiconductor imagery. We treat it as a proxy: the degradation —  $\times 2$  decimation plus signal-dependent noise — is what transfers to inspection imagery, so we characterised the degradation empirically and optimised for degradation robustness rather than fitting content-specific priors.



Measured (D1/D12/D2): downsample kernel is a recovered 4x4 sharpening kernel (bicubic antialias-off is within  $1.22e-05$  of optimal); noise is applied AFTER downsampling, signal-dependent, no additive Gaussian floor (D2, residual autocorrelation  $\sim 0$ ).

Core concept: one-step blind joint restoration, all compute at LR resolution,  $\times 2$  PixelShuffle head -- no cascade, no GAN.

## 4. Proposed Solution

Pipeline: load .npy -> group by shape -> batch -> bf16 forward (channels\_last) -> clip [0,1] -> save

Architecture: NAFSR -- NAFNet-style blocks (SimpleGate, SCA channel attention, LayerNorm) + x2 PixelShuffle head, width 64, 32 blocks, FiLM noise-level condition

Loss: Charbonnier (fidelity) + SSIM (structure) + FFT (frequency) -- balanced, no adversarial term (no hallucination risk for inspection use)

Augmentation: dihedral flips/rotations, CutBlur, on-the-fly synthetic re-degradation with randomised order and noise levels

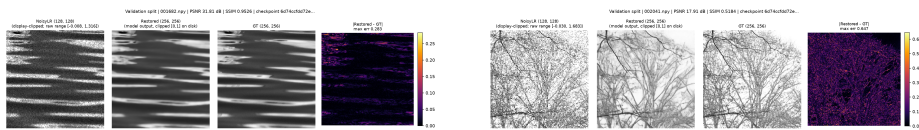
## 5. Innovation & Uniqueness

- (a) Empirical degradation forensics driving a matched synthetic-pair generator -- kernel and noise parameters measured from the data, not assumed from the spec.
- (b) Balanced fidelity + structure + frequency loss with an explicit no-GAN / no-hallucination decision -- justified by inspection semantics: a plausible but wrong structure is worse than a blurry correct one when a die's fate depends on it.
- (c) Throughput-engineered inference path: grouped batching by shape, bf16 + channels\_last, threaded I/O, memory-aware OOM recovery (automatic batch halving, CPU-bicubic floor) -- measured, not assumed.

# 6. Results

| Method                        | PSNR dB            | SSIM                | LPIPS               | n   |
|-------------------------------|--------------------|---------------------|---------------------|-----|
| Bicubic x2 (raw NoisyLR)      | 23.6524 +/- 3.0236 | 0.54775 +/- 0.19197 | 0.41206 +/- 0.15407 | 400 |
| Median 3x3 -> bicubic x2      | 25.5057 +/- 3.8785 | 0.61317 +/- 0.17232 | 0.40870 +/- 0.15866 | 400 |
| Non-local means -> bicubic x2 | 26.2722 +/- 4.3037 | 0.65152 +/- 0.19523 | 0.42586 +/- 0.18627 | 400 |
| U-Net baseline                | 28.8808 +/- 4.5328 | 0.78273 +/- 0.14245 | 0.26525 +/- 0.14878 | 400 |
| Final model                   | 29.5850 +/- 4.6301 | 0.79460 +/- 0.14204 | 0.25416 +/- 0.13263 | 400 |

Proxy-OOD (procedural geometric content, n=40): PSNR 41.4414 +/- 7.1618 / SSIM 0.99691 +/- 0.00307 / LPIPS 0.00169 +/- 0.00172 -- synthetic, not semiconductor imagery; degradation model unchanged, not refit.



Typical case (near-mean PSNR) | Honest failure: worst-PSNR case, fine broadband texture beyond LR recoverability (measured)

## 7. Technology & Feasibility

PyTorch 2.11.0+cu128, CUDA 12.8. Trained on cloud (HF Jobs A100-large GPU);  
inference measured on an NVIDIA RTX 4060 Laptop GPU (8 GB), the dev machine.

Model: NAFSR, 1,393,938 params, checkpoint 11.0 MiB

Inference throughput (128->256, NVIDIA GeForce RTX 4060 Laptop GPU (compute capability 8.9, driver 610.47), bf16, batch 32): 19.40 img/s, fixed startup 30.6% of wall-clock at N=400 -- externally timed, process start to finish

Inference throughput (256->512, NVIDIA GeForce RTX 4060 Laptop GPU (compute capability 8.9, driver 610.47), bf16, batch 32): 11.9 img/s, fixed startup 34.3% of wall-clock at N=400

No H100 number is reported unless explicitly labelled as a projection with its formula shown -- KLA scores on an H100, we trained/measured on an RTX 4060 and (if landed) an A100 cloud run, and never conflate the two.

## 8. GitHub & Video Link

Repository: [https://github.com/sahithsundarw/bunker\\_backer](https://github.com/sahithsundarw/bunker_backer)  
(public, verified in a logged-out window)

Demo video: [[ OPTIONAL, <=5 min, link here if recorded ]]



## 9. References

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